AI-POWERED LEGAL PLATFORM

CHITRAGUPTA

India's Legal First-Response System — Get instant clarity. Know your rights. Take action.

Team BugWeisers

Hackathon Presentation | October 2023

🧠 The Problem

> *80% of Indians cannot afford legal representation. Most don't even know their basic constitutional rights when facing police, landlords, employers, or domestic abuse.*

80%
CAN'T AFFORD A LAWYER

4.7Cr
PENDING COURT CASES

₹0
OUR PRICE TO CITIZENS

💡 Our Solution

Chitragupta is a full-stack, AI-powered Legal First-Response Platform that:

STEP	ACTION	DETAIL
1	**Listens**	Citizen describes their problem in plain Hindi or English
2	**Analyzes**	AI categorizes the legal issue (FIR, consumer fraud, family dispute, etc.)
3	**Guides**	Step-by-step legal roadmaps with IPC sections & jurisdiction-specific advice
4	**Connects**	Matches users with verified lawyers, enables encrypted chat
5	**Protects**	Enterprise-grade 5-layer security for all legal data

✨ Key Features

🤖

AI Legal Triage

📝

RTI Document Generator

NLP-based categorization across 6+ legal domains

Auto-generate Right to Information applications

Lawyer Matching

Connect with verified lawyers by specialization

Encrypted Legal Chat

Real-time chat with end-to-end encryption

Emergency Mode

One-tap helplines: 15100, 112, 181

Bilingual Support

Full Hindi + English language support

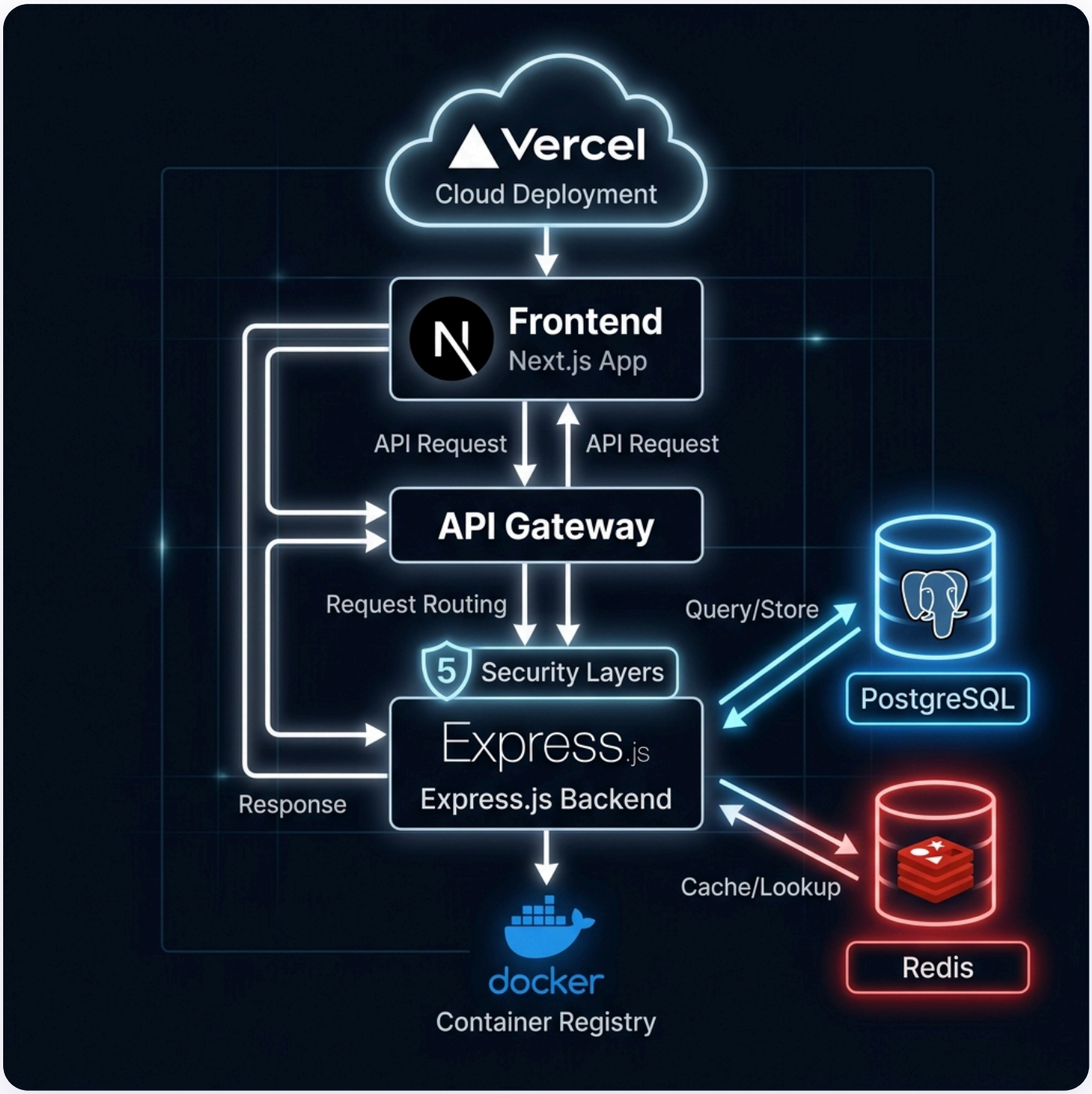
5-Layer Security

WAF, TOTP 2FA, JWT rotation, immutable audit logs

Admin Command Center

Glassmorphic dashboard with CVSS vulnerability scanning

📐 System Architecture



🛠️ Tech Stack

LAYER	TECHNOLOGY	PURPOSE
Frontend	Next.js 14	SSR, App Router, SEO

Styling	Tailwind CSS v3	Utility-first responsive design
Auth	Firebase Auth	OAuth (GitHub), token mgmt
Backend	Express.js 5	REST API server
Database	PostgreSQL 15	Relational data, audit logs
Cache	Redis 7	Rate limiting, sessions
Security	Helmet, HPP, XSS-Clean	Request hardening
2FA	Speakeasy (TOTP)	Google Authenticator
Logging	Winston + Rotate	Structured audit trail
Deploy	Vercel + Docker	CI/CD + orchestration

🔗 API Endpoints

Authentication (Layer 1 & 2)

METHOD	ENDPOINT	DESCRIPTION
POST	`/api/auth/register`	Register with OTP verification
POST	`/api/auth/verify-otp`	Verify 6-digit OTP code
POST	`/api/auth/login`	Login (returns JWT pair)
POST	`/api/auth/refresh`	Rotate JWT tokens
POST	`/api/auth/totp/setup`	Generate TOTP QR for 2FA

Admin Security (Layer 4 & 5)

METHOD	ENDPOINT	DESCRIPTION
GET	`/api/admin/logs`	Fetch immutable audit logs
GET	`/api/admin/logs/export/csv`	Export logs as CSV

GET	/api/admin/logs/export/pdf	Export logs as PDF
POST	/api/admin/scanner/run	Trigger vulnerability scan
GET	/api/admin/scanner/reports	Fetch scan reports

📂 Project Structure

```
Chitragupt/
├── frontend/                    # Next.js 14 Application
│   ├── app/
│   │   ├── admin/               # 🛡️ Security Admin Dashboard
│   │   ├── chat/                # 💬 Encrypted Legal Chat
│   │   ├── dashboard/           # 📊 User Dashboard
│   │   ├── documents/           # 📝 RTI Generator
│   │   ├── get-help/            # 🚨 AI Legal Triage
│   │   ├── lawyers/             # 👨‍⚖️ Lawyer Directory
│   │   ├── login/               # 🔐 Authentication
│   │   ├── my-case/             # 📋 Case Management
│   │   └── register/            # ✍️ Registration
│   └── components/
│       ├── admin/               # Admin Panel (AuditLogTable, Scanner)
│       ├── auth/                # Auth Components
│       └── ui/                  # Shared UI (Navbar, Footer, FAB)
│
├── backend/                     # Express.js 5.x API
│   └── src/
│       ├── config/              # DB, Redis, Firebase, Email, Twilio
│       ├── database/migrations/ # SQL Schema
│       ├── middleware/          # WAF, Auth, Rate Limit, Audit
│       ├── modules/
│       │   ├── auth/            # Register, Login, Logout, Refresh
│       │   ├── otp/             # 6-digit OTP (Email + SMS)
│       │   ├── totp/            # TOTP 2FA (Google Authenticator)
│       │   ├── audit/           # Log Viewer + CSV/PDF Export
│       │   ├── notifications/   # FCM + Web Push Alerts
│       │   └── scanner/         # Vulnerability Self-Scanner
│       └── jobs/                # Scheduled Security Scan
│
└── docker-compose.yml           # PostgreSQL 15 + Redis 7
```

🏆 Why Chitragupta Wins

CRITERIA	OUR APPROACH
Innovation	First AI-powered legal triage designed specifically for Indian law
Impact	Addresses access-to-justice gap affecting 1B+ citizens

Security	5-layer defense-in-depth — beyond typical hackathon prototypes
Completeness	Full-stack: Frontend + Backend + DB + Auth + Admin + Deployment
Scalability	Containerized with Docker, deployed on Vercel edge network
Bilingual	Native Hindi + English support for true inclusivity